

Priyadharshini Ganeshkumar

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Summary

Full-stack software engineer with an M.S. in Computer Science from UC Davis and 3+ years shipping AI-native products and scalable backend services. Builds LLM and agentic systems into real user workflows end to end, from Next.js and React frontends to Python backends, retrieval pipelines, and evaluation. Experienced in fast-moving startups, iterating from customer feedback to production, with a grounding in reliability, security, and multi-tenant data isolation for high-trust domains.

Experience

Software Engineer | Nyck AI, USA

Jun 2026 - Present

AI-native B2B procurement and operations platform (early-stage startup)

- Built a secure **client-facing conversational AI interface** on a decoupled **FastAPI** agentic layer exposing **GraphQL and REST** endpoints, translating ambiguous customer requests into **agentic actions** across core procurement and inventory workflows.
- Forward-deployed** with enterprise clients, shipping full-stack features **0 to 1** from feedback to production; built a **Notion MCP** loop that auto-triages client tickets and dispatches **AI coding agents** to draft fixes, optimizing the feedback-to-resolution cycle.
- Architected **multi-tenant data isolation** with **Supabase Row-Level Security**, enforcing per-client data boundaries for a privacy-sensitive B2B product; built ERP ingestion pipelines into **Postgres** that cut new-customer onboarding time by **70%**.

Software Engineer | JPMorgan Chase, India

Jan 2022 - Jul 2024

- Designed and shipped trade reconciliation systems for **10+ enterprise clients** on Linux batch infrastructure, migrated HDFS batch workflows to **Amazon S3 and Spark**, improving reconciliation time by **80%**.
- Partnered with customer stakeholders and development teams to gather batch failure-detection requirements and built a serverless **AWS solutions** using **Lambda, CloudWatch, SNS and S3**, improving reliability and ensuring **100%** SLA compliance.
- Migrating legacy database workflows to **Amazon DynamoDB & Oracle GOS**, optimizing and refactoring complex PL/SQL to improve query performance by **30%** and unblock **20+ downstream applications**.
- Deployed and operated **containerized microservices at scale** using **Docker and Kubernetes**, supporting high-throughput distributed & parallel financial workloads; managed capacity planning and SLA enforcements across cloud infrastructure.
- Detected privilege escalation, anomalous database access and payment flow risks in real time; integrated with **SIEM/SOC tooling** for **60% faster incident response** with zero production performance impact on high-throughput bare-metal Linux workloads.
- Supported production operations in an **on-call capacity**, leading post-migration game-day testing and failure-mode runbook for pod crashes, traffic surges, API timeouts and autoscaling events to validate recovery and availability.

Software Engineer Intern | JPMorgan Chase, India

May 2021 - Jul 2021

- Modernized legacy **ETL data pipeline** by decommissioning Informatica and migrating to **SQL Loader**; automated regression testing using maintainable Python scripts to ensure zero-risk cutover, reducing operational cost by **\$50,300 annually**.
- Integrated **CI/CD** pipelines using **Jenkins, SonarQube and Control-M**, enforcing code quality gates, automated testing and secure release processes across the SDLC, reducing deployment cycle time by **30 minutes**.

Projects

SiteSafe AI Inspector - Vertical AI Agent for Google DeepMind Hackathon | Gemini, LoRA, Fine-tuning [Github](#) Feb 2026

- Built an AI agent mapping multi-modal inputs to a 70+ item safety checklist, auto-generating audit reports. Fine-tuned Gemma-3-4B IT via LoRA on NVIDIA RTX 6000 GPU for domain-adapted visual hazard recognition, eliminating manual overhead.

Deep Researcher - Multi Agent LLM Pipeline | CrewAI, MCP, RAG, Gradio, Serper [Github](#) [Demo](#) Dec 2025

- Built an evaluator-optimizer agentic workflow for autonomous **literature summarization** with **semantic search, retrieval & relevance ranking** over multi-source results; designed AI evaluation loop achieving **80%** factual alignment with human reviews.

Accelerated PBFT Consensus via RDMA | zRPC, ResilientDB [Github](#) Mar 2026

- Implemented RDMA Shared/Per-Client Ring buffers to replace TCP/IP in PBFT consensus; achieved 35% lower commit latency and 40% higher throughput in multi-node distributed setups.

Education

University of California, Davis | M.S. Computer Science (GPA 3.9/4.0)

Sep 2024 - Jun 2026

Coursework: Machine Learning & Agentic AI, Information Security, Software Engineering, Business Analytics (Teaching Assistant)

PSG College of Technology, India | B.Tech Information Technology (GPA 3.8/4.0)

Aug 2018 - Mar 2022

Coursework: Object-Oriented Programming, Data Structures and Algorithms, Network Security, Distributed Database Systems.

Skills & Awards

Technical Skills: AI/ML: Agentic AI, LLM Integration & Orchestration, RAG, MCP, LangChain, CrewAI, AutoGen, Evaluation Workflows, Claude Code, Codex, **Full-Stack:** React, Next.js, TypeScript, JavaScript, Node.js, FastAPI, REST APIs, GraphQL, gRPC, **Languages:** Python, Java, Go, C, C++, PL/SQL, BASH, **AWS:** Lambda, S3, RDS, EKS, UI/UX, IAM, **Databases:** PostgreSQL, Supabase, Redis, Oracle, DynamoDB (SQL & NoSQL), **Security:** RLS, TLS/SSL, WAF, SIEM, Linux/Unix

Certifications: AWS Certified Solutions Architect - Associate, AWS Certified Cloud Practitioner.

Awards: Recognized in Q1 Appreciation Scroll at JPMC for outstanding contributions to migration project and mentoring interns.